PeakWatch

Methodology & data resource guide — Marek Krawczyk · July 2026 ·
github.com/MarekSearchingForData/peakwatch

Rule of the project: **every layer reconciles against an independent published number before anything is built on it.**

Data resources

Resource	What it is	How we use it	Gotchas
ISO-NE Web Services — combinedhourlydemand	Day-ahead cleared (DaLoad) and actual (RtLoad) demand per load zone, hourly, 2022–present	Core load history; ISO's DA doubles as the benchmark forecast	Free account; aggressive 429 rate limits (sequential + backoff); RtLoad settles days late; location IDs verified 4001=ME … 4008=NEMA
ISO-NE Monthly Regional Network Load reports	Settlement-grade MW of every network customer at the monthly transmission peak hour	Ground truth for town loads; model anchor and scoring target (20 towns × 49 months)	~2-month lag; restated (-drp) files supersede; zero values are real (own generation); a 15× first-publication typo has occurred
ISO-NE fiveminutesystemload / hourlyloadforecast	Live system load + ISO's own hourly forecast	Dashboard live tiles; forecast feature and benchmark	5-min feed is system-wide only; zone actuals lag ~2 days
Meteostat	Station-observed hourly weather per town, 2023–present	Observed-weather history	Station gaps; pinned to v1.x
Open-Meteo (archive + forecast)	Modeled hourly weather incl. solar radiation (GHI), cloud, wind; 2022–present + 7-day forecast	Offset features (rooftop PV, wind) — same source family for training and live serving	Modeled, not observed; legacy variable names; per-minute rate cap
EIA Forms 860 / 861 + MassCEC	Generator inventory per town; utility sales by customer class	Town portfolios (86 assets, status-flagged) and class-mix weights; annual-energy validation constraint	Annual cadence; 14 of 20 munis file the short form (totals only)
ISO-NE OATT / FCA results	RNS transmission rate ($183.71/kW-yr 2026); capacity clearing prices (FCA15–18)	Dollarization: $105.5M/yr membership exposure; $/MW-shaved	Capacity year runs June–May; a peak sets tags for the FOLLOWING June's year
MassGIS	Town boundary shapefile (351 towns)	Interactive dashboard map	Already WGS84; simplified to 0.5 MB

Definitions

- **RNL** (Regional Network Load): a customer's load at the hour of the monthly transmission (RNS) peak. Sets transmission charges — twelve bill-setting hours per year.
- **Capacity tag**: load at the single annual ISO-NE coincident peak hour; sets capacity charges for the commitment year starting the following June.
- **Share (alpha)**: town RNL ÷ zone load at the settlement anchor hour.
- **Anchor snapping**: the settlement hour is identified as the hour whose summed-zone load best matches the published pool value (±0.84% mean gap).

Modeling protocol

- **Champion/challenger zoo.** Every model is re-scored on every run via leave-one-month-out MAPE (absolute MW for micro towns) against settlement truth; the per-town champion is the current winner; all scores persist in the scorecard. Challengers are never deleted.
- **Features earn admission.** A signal ships only if it beats the incumbent out-of-sample. Physics guides candidates (GHI works through rooftop-PV offset); the scorecard decides.
- **Honest scoring only.** Evaluations use information available at prediction time; walk-forward re-tests confirm the LOO rankings are not leakage artifacts.
- **Consensus.** Top-3 model averaging is itself a scored challenger (current champion for Holyoke: 6.99% vs 7.83% flat).
- **Uncertainty ships with every number.** Bands from out-of-sample residual simulation, calibrated to the 80% target.
- **Predict-before-publish.** Monthly town predictions are sealed in version control before ISO releases the truth (~2-month lag) and scored on arrival. July 2026 predictions for all 20 towns were sealed 2026-07-22.

Measured results (as of 2026-07-23)

- Peak-day capture: 95.5% of monthly peaks at 6.8 alert-days/month (ML model, 22-month sample); 100% at 14.8 alert-days on 54 months (transparent rule). Held-out July 2026: alerted on the annual peak day (25,321 MW, July 2 at 6 PM) with zero false alarms after.
- Zone forecast: LightGBM quantile model blended 50/50 with ISO's day-ahead beats ISO alone on every window tested (e.g. 2.80% vs 3.81% MAPE).
- Town settlement prediction: Chicopee 3.53% (temp+GHI), Holyoke 6.99% (consensus), Princeton ±0.72 MW (temp+wind) — each town's champion matches its physical situation.
- Reference points: VPPSA contracts battery dispatch at 90% capture; Enel X issues ~6–9 alert days per summer.

Validation gates (run on every refresh)

Zone magnitude ranking (CT largest, VT smallest — label-swap detector) · DA↔RT correlation per Massachusetts zone · day completeness · settlement regions match the verified town-zone mapping · RNL non-negative (zero allowed — own generation) · spike guard (flags any town-month over 4× its own median) · promotion to clean tables only when all blocking checks pass, else quarantined.

Known limitations

- Generation-heavy towns (Princeton class) remain hardest; hub-height wind and true turbine power curves are queued refinements.
- Settlement reports before mid-2022 not yet located.
- Capacity tags proxied by latest summer settlement values until actual ICAP tags are available.
- Cloud deployment runs on a committed data snapshot; live tiles are real-time, history updates with each snapshot push.